Day 38 EBIKE101 Morning Quiz — Instructor Answer Key

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Topic: E-bike Homunculous diagnosis and finish-work planning

Answer Key with Rationales

1. **B** — Diagnosis starts with safety. Students should confirm the system is safe to inspect and de-energized when appropriate before touching unknown wiring.

2. **A** — An unfinished build needs inventory first: what exists, what is missing, what is loose, what is unknown, and what must be verified.

3. **B** — A fuse or breaker protects wiring and parts from unsafe current. It is not a performance part.

4. **B** — A dead display can come from missing battery voltage, switch/fuse issues, or a disconnected display/controller path.

5. **B** — A missing throttle is a missing control input. It does not automatically prove the system is impossible, but it must be documented and checked against the controller/display/PAS behavior.

6. **B** — Wire color can be a clue, but e-bike systems often involve adapters, repairs, and mixed manufacturers. Testing beats color-confidence theater.

7. **A** — Continuity testing belongs on a de-energized circuit or component. Using continuity mode on a live battery circuit can damage the meter or create unsafe conditions.

8. **A** — Brake cutoffs tell the controller not to drive the motor. A stuck or misadjusted cutoff can make a working system appear dead.

9. **A** — PAS usually means Pedal Assist Sensor/System. It detects pedaling input so the controller can assist.

10. **B** — Strain relief protects wires from pulling, rubbing, bending, and wheel movement.

11. **B** — A blown fuse is a symptom. Before replacement, inspect for shorts, wrong connections, pinched wires, or overload.

12. **B** — Tracing means following the real path from source, through protection/control, to load.

13. **B** — Powered testing requires mechanical safety first: secure wheel, axle/torque support, brakes, chain path, and clear moving parts.

14. **B** — First powered testing should be controlled, low-power, secured, and instructor-led. No shop test rides.

15. **B** — A useful diagnostic log captures symptoms, observations, measurements, changes, parts needed, and next step.

16. **A** — The controller turns rider/display inputs into controlled electrical power for the motor.

17. **B** — Heat, burnt smell, or melted wiring means stop and get instructor review. More testing can make damage or fire risk worse.

18. **A** — Connectors fail in boring ways: bent pins, loose plugs, corrosion, bad matches, or damaged wires. Naturally, boring failures create dramatic symptoms.

19. **B** — A symptom is what is observed. A root cause is the underlying reason.

20. **B** — “Finished” means verified mechanical safety, electrical safety, control function, supported wiring, known missing parts, and documentation.

Suggested grading

- 18–20: Ready for diagnostic leadership role today.
- 15–17: Solid working understanding; pair with peer verification.
- 12–14: Review safety and system tracing before powered work.
- Below 12: Keep student in observation/documentation/mechanical-support role until reteach.

Reteach focus

If many students miss:

- Questions 1, 7, 14, 17: reteach safety boundaries before lab.
- Questions 5, 8, 9, 16: reteach control inputs and controller logic.
- Questions 11, 12, 15, 19: reteach troubleshooting process and documentation.